

**SIMATS ENGINEERING
ASSESSMENT TOOL |**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

STUDENT NAME: PRIYANKA . G

REG. NO : 192524124 .

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 4

Time Duration: 1 Hour

Total Marks: 40

Annexure A

CASE STUDY BASED QUESTIONS

Code of Conduct:

I Priyanka G / 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Priyanka G

25+25+20+19+9

98

PG

Annexure A

CASE STUDY

Student 31: Priyanka G | Register Number: 192524124

Question 1: Case Study - Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
 - II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
 - III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
 - IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True
- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Question 2: Case Study - Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. for-all x: Vehicle(x) AND EmergencyType(x) $\rightarrow$ ClearPath(x) [Rule 1]
 - II. for-all x: ClearPath(x) $\rightarrow$ GreenSignal(x) AND NOT RedSignal(x) [Rule 2]
 - III. for-all x for-all y: Vehicle(x) AND Behind(x,y) AND GreenSignal(y) $\rightarrow$ Proceed(x) [Rule 3]
 - IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)
- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set theta (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed(CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Question 3: Case Study - Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
 - 2) P2: IrrigationNeeded AND CropWheat $\rightarrow$ ApplyDripMethod
 - 3) P3: NOT ApplyDripMethod AND CropWheat $\rightarrow$ CropAtRisk
 - 4) Facts: SoilDry = True, CropWheat = True
- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move NOT inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Question 4: Case Study - Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench[1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze[1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter[x,y] $\rightarrow$ Gold is at [x,y]
- 4) NOT Wumpus[x,y] AND NOT Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

DATE: 13/08/26

CSAIFIA ARTIFICIAL INTELLIGENCE

NAME: PRIYANKA G

COS ASSESSMENT TOOL 1

REG. NO: 192524124

CASE STUDY

1) SMART MEDICAL DIAGNOSIS SYSTEM

(a) Propositional Logic Representation

Symbol	Meaning
F	Patient A has Fever
C	Patient A has Cough
R	Patient A has Rash
B	Patient A has Breathlessness
Flu	Patient A has possible Flu
M	Patient A has possible Measles
P	Patient A has possible Pneumonia

Rules:

- Rule I: $F \wedge C \rightarrow \text{Flu}$
- Rule II: $F \wedge R \rightarrow M$
- Rule III: $C \wedge B \rightarrow P$

Facts:

F = True

C = True

B = True

There is no fact stating $R = \text{True}$, so from that we can conclude that measles rule cannot be fired.

(b) Forward Chaining

Forward Chaining starts with known facts and repeatedly applies rules whose antecedents are satisfied.

⇒ Initial Facts : $KB_0 = \{F, C, B\}$

Step 1: Apply Rule I (i.e. $FAC \rightarrow Flu$)

∵ $F = \text{True}$, $C = \text{True}$ ∴ $Flu = \text{True}$

⇒ Derived fact : Patient A may have Flu.

Step 2: Apply Rule II (i.e. $FAR \rightarrow M$)

∵ $F = \text{True}$, but $R = \text{False / Unknown}$

∴ Rule II cannot fire. So, $M \notin KB$

Step 3: Apply Rule III (i.e. $CAB \rightarrow P$)

∵ $C = \text{True}$, $B = \text{True}$ ∴ $P = \text{True}$

⇒ Derived fact : Patient A may have Pneumonia.

Final knowledge: $\{F, C, B, Flu, P\}$

Final Diagnosis: Thus, the system identifies Flu and Pneumonia as possible diagnoses from the available facts.

(c) Backward Chaining for Pneumonia:

Goal : Goal = P , we work backward from the desired conclusion.

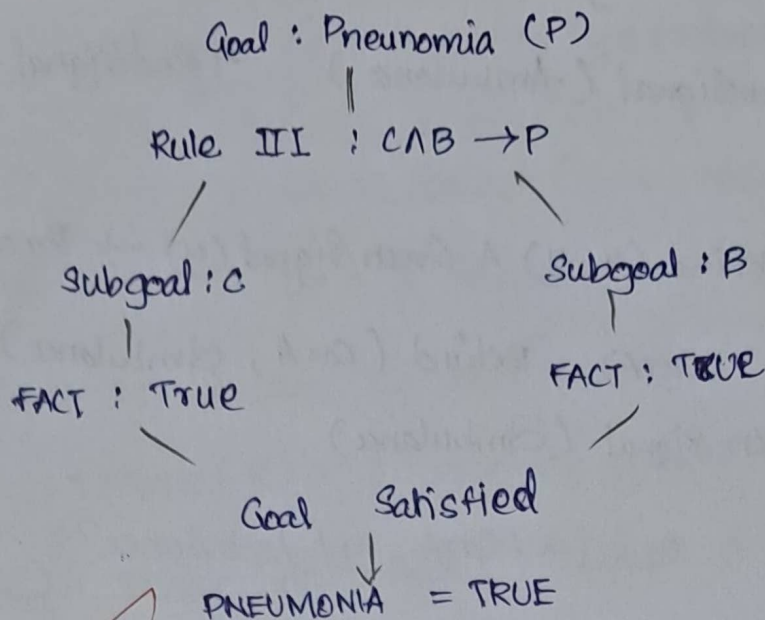
The only rule that concludes P is: $CAB \rightarrow P$

∴ The goal P is reduced to two subgoals: C and B

Since both C and B are known facts;

$C = \text{True} \wedge B = \text{True}$

∴ $P = \text{True}$



Conclusion: Patient A has possible Pneumonia according to the rule-based knowledge base.

2) AUTONOMOUS TRAFFIC MANAGEMENT AGENT.

(a) Unification

Rule I $\forall x \text{ Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{clearPath}(x)$

Given facts: $\text{Vehicle}(\text{Ambulance})$
 $\text{EmergencyType}(\text{Ambulance})$

Now, we need to match $\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance})$

$$\therefore \theta_1 = \{x / \text{Ambulance}\}$$

Now, substitute $x = \text{Ambulance}$:

$\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance}) \rightarrow \text{clearPath}(\text{Ambulance})$

Now, both premises are present, therefore $\text{clearPath}(\text{Ambulance})$

MGU: The Most General Unifier is: $\theta = \{x / \text{Ambulance}\}$

Applying Rule II: $\text{clearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$

Let's substitute, $x = \text{Ambulance}$

$$\therefore \text{clearPath}(\text{Ambulance}) \rightarrow \text{GreenSignal}(\text{Ambulance})$$

and ClearPath (Ambulance) $\rightarrow \neg$ RedSignal (Ambulance)

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Hence: GreenSignal (Ambulance) $\neg$ RedSignal (Ambulance)

Applying Rule iii:

Vehicle (x) $\wedge$ Behind (x, y) $\wedge$ GreenSignal (y) $\rightarrow$ Proceed (x)

Given: Vehicle (CarA), Behind (CarA, Ambulance)
GreenSignal (Ambulance)

Unification gives: $\theta_2 = \{x / \text{CarA}, y / \text{Ambulance}\}$

$\therefore$ Proceed (CarA)

(b) Resolution Proof

To Prove: Proceed (CarA)

Resolution uses refutation, so negate the goal: $\neg$ Proceed (CarA)

CNF Clauses:

Rule 1: Vehicle (x) $\wedge$ EmergencyType (x) $\rightarrow$ ClearPath (x)

Eliminate Implication:

$\neg$ Vehicle (x) $\vee \neg$ EmergencyType (x) $\vee$ ClearPath (x)

clause:

$C_1 = \neg$ Vehicle (x) $\vee \neg$ EmergencyType (x) $\vee$ ClearPath (x)

Rule 2: ClearPath (x) $\rightarrow$ GreenSignal (x)

$C_2 = \neg$ ClearPath (x) $\vee$ GreenSignal (x)

The negative red-signal conclusion is

$C_3 = \neg$ ClearPath (x) $\vee \neg$ RedSignal (x)

Rule 3: Vehicle (x) $\wedge$ Behind (x, y) $\wedge$ GreenSignal (y) $\rightarrow$ Proceed (x)

CNF: $C_4 = \neg$ Vehicle (x) $\vee \neg$ Behind (x, y) $\vee \neg$ GreenSignal (y) $\vee$ Proceed (x)

Facts:

$C_5 = \text{Vehicle (Ambulance)}$

$C_6 = \text{Emergency Type (Ambulance)}$

$C_7 = \text{Vehicle (Car A)}$

$C_8 = \text{Behind (Car A, Ambulance)}$

Negated Goal:

$C_9 = \neg \text{Proceed (Car A)}$

Resolution Step 1: Resolve C_1 with C_5

$\neg \text{Vehicle}(x) \vee \neg \text{Emergency Type}(x) \vee \text{ClearPath}(x)$
and $\text{Vehicle}(\text{Ambulance})$ Using: $\theta = \{x / \text{Ambulance}\}$
 $\Rightarrow \neg \text{Emergency Type}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$

Resolution Step 2: Resolve with $\text{Emergency Type}(\text{Ambulance})$

$\therefore \text{ClearPath}(\text{Ambulance})$

Resolution Step 3: Using $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

and $\text{ClearPath}(\text{Ambulance})$

$\Rightarrow \text{GreenSignal}(\text{Ambulance})$

Resolution Step 4: Use Rule III:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

with $\text{Vehicle}(\text{Car A})$ and $\text{Behind}(\text{Car A, Ambulance})$ and $\text{GreenSignal}(\text{Ambulance})$

$\rightarrow$ Substitution, $\theta = \{x / \text{Car A}, y / \text{Ambulance}\}$

$\therefore \text{Proceed}(\text{Car A})$

Resolution Step 5: we have $\text{Proceed}(\text{Car A})$ and the negated goal $\neg \text{Proceed}(\text{Car A})$

$\Rightarrow \text{Proceed}(\text{Car A}), \neg \text{Proceed}(\text{Car A})$

gives $\square$, where $\square$ is the empty clause.

Conclusion:

$\therefore$ Empty clause is derived: $\text{KB} \models \text{Proceed}(\text{Car A})$

$\therefore$ Car A can proceed behind Ambulance based on knowledge

(C) Two Simultaneous Emergency Vehicle.

The current KB is completely sufficient for safely handling two simultaneous emergency vehicles.

Suppose Vehicle (Ambulance 1), EmergencyType (Ambulance 1)
and Vehicle (Ambulance 2), EmergencyType (Ambulance 2).

Rule 1 can independently derive: ClearPath (Ambulance 1) and
ClearPath (Ambulance 2).

However, the KB doesn't contain for:

- which emergency vehicle receives priority.
- what happens when both require the same road.
- whether two emergency vehicles can simultaneously receive green signals.
- Conflict resolution between emergency routes.
- Maintaining safe distance between emergency vehicles.

Additional Rule Needed

A priority rule could be introduced:

EmergencyPriority(x,y) $\rightarrow$ PriorityVehicle(x)

A conflict could be introduced:

SameRoute(x,y) $\wedge$ Priority(x,y) $\rightarrow$ Hold(x)

A resource-conflict rule could be:

GreenSignal(x) $\wedge$ GreenSignal(y) $\wedge$ Conflict(x,y) $\rightarrow$ PriorityDecision(x,y)

Logical Justification

The existing KB derives individual permission, but it does not represent interaction between multiple emergency vehicles.

$\therefore$ Current KB is insufficient for simultaneous emergency-vehicle conflict handling.

AGRICULTURAL EXPERT REASONING SYSTEM

(a) CNF Conversion

Given: P_1 : Soil Dry $\rightarrow$ Irrigation Needed

$\Rightarrow$

Eliminate Implication: $\neg$ Soil Dry $\vee$ Irrigation Needed

$$\therefore P_1 = \neg \text{Soil Dry} \vee \text{Irrigation Needed}$$

$\Rightarrow P_2$: Irrigation Needed $\wedge$ CropWheat $\rightarrow$ ApplyDripMethod

Eliminate Implication: $\neg(\text{Irrigation Needed} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

Apply DeMorgan's law:

$$\therefore P_2 = \neg \text{Irrigation Needed} \vee \neg \text{Crop Wheat} \vee \text{ApplyDripMethod}$$

$\Rightarrow P_3$: $\neg$ ApplyDripMethod $\wedge$ Crop wheat $\rightarrow$ CropAtRisk

$$\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$$

$$\therefore P_3 = \text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$$

Facts: Soil Dry = True

CNF: Soil Dry

and: CropWheat = True

CNF: CropWheat

Final CNF Knowledge Base:

$$C_1 = \neg \text{Soil Dry} \vee \text{Irrigation Needed}$$

$$C_2 = \neg \text{Irrigation Needed} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$$

$$C_3 = \text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$$

$$C_4 = \text{Soil Dry}$$

$$C_5 = \text{CropWheat}$$

(b) Resolution Refutation for ApplyDripMethod.

Theorem: ApplyDripMethod $\Rightarrow$ Negated Goal: $\neg$ ApplyDripMethod

Resolution 1: Take $C_1 = \neg$ SoilDry $\vee$ IrrigationNeeded and SoilDry
therefore, IrrigationNeeded.

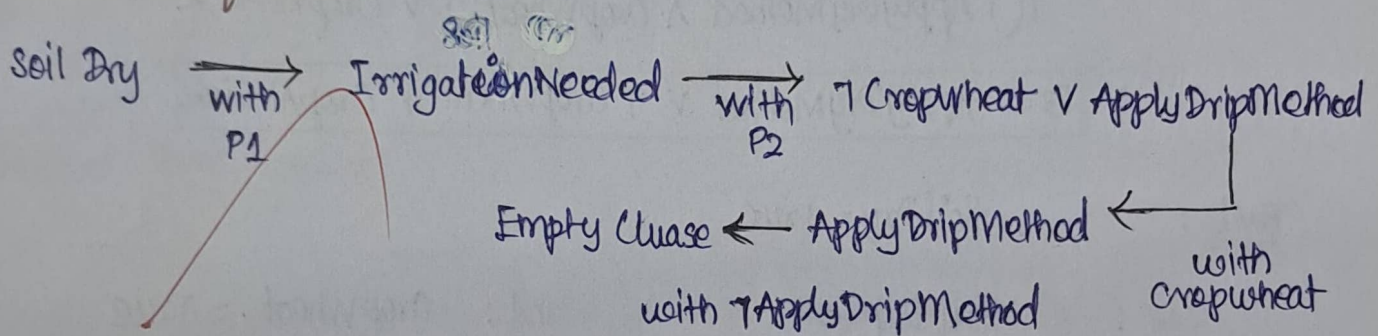
Resolution 2: $C_2 = \neg$ IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod
and Irrigation Needed, $\therefore \neg$ CropWheat $\vee$ ApplyDripMethod.

Resolution 3: Use CropWheat, Resolve with $\neg$ CropWheat $\vee$ ApplyDripMethod
 $\therefore$ ApplyDripMethod.

Resolution 4: $\neg$ ApplyDripMethod, ApplyDripMethod (from resolution)

Result: $\square$

Resolution Proof Tree:



$\therefore$ KB $\neq$ ApplyDripMethod

Conclusion: The agriculture system logically proves that the Drip Irrigation Method should be applied.

(c) Removing SoilDry

Originally: Soil Dry = True was responsible for deriving IrrigationNeeded. and Subsequently: ApplyDripMethod

If SoilDry is removed, the KB becomes;

$$C_1 = \neg$$
SoilDry $\vee$ Irrigation Needed

$$C_2 = \neg$$
IrrigationNeeded $\vee$ $\neg$ CropWheat $\vee$ ApplyDripMethod

$$C_3 = \text{ApplyDripMethod} \vee \neg$$
CropWheat $\vee$ CropAtRisk

there is now no fact establishing SoilDry $\Rightarrow$ CropWheat

ApplyDripMethod

IrrigationNeeded cannot be derived through P1, without IrrigationNeeded P2 cannot establish: ApplyDripMethod

∴ we cannot derive $\neg$ ApplyDripMethod.

∴ P3 is $\neg$ ApplyDripMethod $\wedge$ Cropwheat $\rightarrow$ CropAtRisk

For CropAtRisk, we require, $\neg$ ApplyDripMethod and Cropwheat

Although: Cropwheat = True is still known, there is no evidence that $\neg$ ApplyDripMethod is True.

∴ CropAtRisk does not follow.

So in classic logic, failure to prove: ApplyDripMethod $\neq$ $\neg$ ApplyDripMethod

Hence, the correct conclusion is unknown, not "CropAtRisk = True"

Futal Analysis:

Condition	Original KB	Soil Dry Removed
Soil Dry	True	Unknown
IrrigationNeeded	Derived	Not Derived
Cropwheat	True	True
ApplyDripMethod	Derived	Not Derived
$\neg$ ApplyDripMethod	Not Derived	Not Derived
CropAtRisk	Not Derived	Not Derived.

Thus, Removing Soil Dry breaks the inference chain, but doesn't imply CropAtRisk.

4. WUMPUS WORLD KNOWLEDGE AGENT.

(a) Belief state and Modus Ponens

Initial Percepts: The agent perceives

Stench [1,2], Breeze [1,1], Gutter [2,2]

∴ Initial belief state: $B_0 = \{ \text{Stench [1,2], Breeze [1,1], Gutter [2,2]} \}$

Rule 1: $\text{Stench}[1,2] \rightarrow \text{Wumpus Adjacent}[1,2]$

Given: $\text{Stench}[1,2]$

By Modus Ponens: $\text{Wumpus Adjacent}[1,2]$

Rule 2: $\text{Breeze}[1,1] \rightarrow \text{Pit Adjacent}[1,1]$

Given: $\text{Breeze}[1,1]$

By Modus Ponens: $\text{Pit Adjacent}[1,1]$

Rule 3: $\text{Glitter}[x,y] \rightarrow \text{Gold}[x,y]$

Given: $\text{Glitter}[2,2]$

By Modus ponens: $\text{Gold}[2,2]$ substitution $\theta = \{x/2, y/2\}$

$\therefore \text{Gold}[2,2]$.

Rule 4: $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

However, the agent has not established: $\neg \text{Wumpus}[x,y]$
and $\neg \text{Pit}[x,y]$ for a specific cell..

$\therefore$ Rule 4 cannot yet be fired.

Derived knowledge:

$\text{Stench}[1,2]$

$\text{Breeze}[1,1]$

$\text{Glitter}[2,2]$

$\text{Wumpus Adjacent}[1,2]$

$\text{Pit Adjacent}[1,1]$

$\text{Gold}[2,2]$

From the derived knowledge base we can conclude that the $\text{Wumpus Adjacent}[1,2]$, $\text{Pit Adjacent}[1,1]$ and $\text{Gold}[2,2]$ from the rules given and the implication rule we applied to obtain them as a premises.

forward Chaining - Wumpus and Pit Locations

The important point is that the rules supplied in the question provide adjacency information, not an exact unique coordinate.

step 1 - Stench

Given: stench $[1,2]$

Rule: stench $[1,2] \rightarrow$ WumpusAdjacent $[1,2]$

By Modus Ponens: $\therefore$ WumpusAdjacent $[1,2]$

The Wumpus must therefore be in cell adjacent to $[1,2]$.
Under the standard four-directional Wumpus world assumption,
candidate neighboring cells are: $[1,1], [1,3], [2,2]$

so, Wumpus $\in \{[1,1], [1,3], [2,2]\}$

Subject to the actual board boundaries.

step 2 - Breeze

Given: Breeze $[1,1]$.

Rule: Breeze $[1,1] \rightarrow$ PitAdjacent $[1,1]$

Therefore: PitAdjacent $[1,1]$

Under four-directional adjacency, possible pit cells are: $[1,2], [2,1]$

$\therefore$ Pit $\in \{[1,2], [2,1]\}$

step 3 - Glitter

Given: Glitter $[2,2]$

Rule: Glitter $[2,2] \rightarrow$ Gold $[2,2]$

Therefore: Gold $[2,2]$

Logical Limitation: The supplied KB does not contain rules such as:

Stench $[x,y] \leftrightarrow$ WumpusAdjacent $[x,y]$

nor rules that eliminate individual neighbouring cells.

Therefore it is logically incorrect to claim a unique Wumpus location from the supplied rules alone.

The strongest valid conclusion is : Wumpus is adjacent to [1,2]
and Pit is adjacent to [1,1]

(c) Safety of Path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$

We need to evaluate each cell.

Cell [1,1]: The agent has : Breeze [1,1] and $\therefore$ Pit Adjacent [1,1]

This indicates a possible pit in a neighboring cell. However, the rule set does not state that [1,1] itself contains a pit. Therefore, [1,1] is not proven unsafe, but KB does not establish complete safety either.

Cell [1,2]: The agent perceives: Stench [1,2] $\therefore$ Wumpus Adjacent [1,2]

Hence, there may be a Wumpus in one of the neighbouring cells. The supplied KB does not prove : $\neg$ Wumpus [1,2] or $\neg$ Pit [1,2]. Therefore [1,2] cannot be classified as definitely safe.

Cell [2,2]: The agent perceives: Glitter [2,2] $\therefore$ Gold [2,2]

so, the gold is confirmed to be at : [2,2]

But the KB does not establish : $\neg$ Wumpus [2,2] and $\neg$ Pit [2,2]

$\therefore$ The safe rule : $\neg$ Wumpus[x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]
can't be applied to [2,2].

Final safety decision: The path : $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ can't be proven from the supplied KB. This does not mean the path is definitely dangerous. It means the agent has insufficient information to establish safety. Path Safety = INCONCLUSIVE.

The logically safest conclusion is therefore : The agent should not classify the complete path as safe because the KB doesn't provide sufficient negative decision to prove that traversal cells contains them.